



Higher Diploma in Computer Science

Computer Systems & Networks



Contact Me

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Or

Tag my name on slack
#caroline cahill



HDIP 25

Semester 2
2025

Computer Systems & Networks



logic · computer
organisation · os · networks
· interfaces · sensors

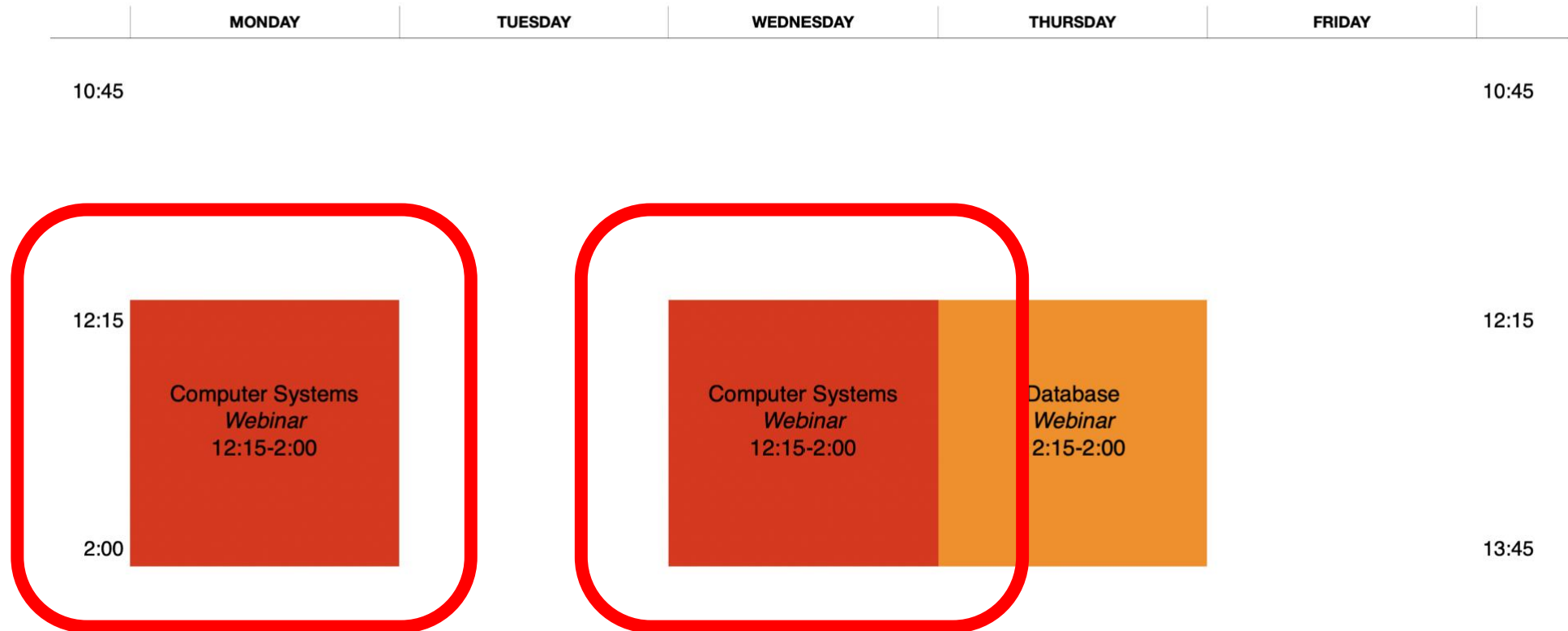
Database



entities · tables · rows · sql
· er · nosql

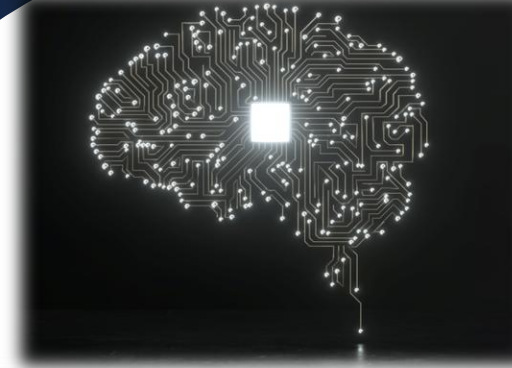
2025		S	M	T	W	T	F	S	Modules
Semester 2									
September									
	1		1	2	3	4	5	6	comp sys & database
	2	7	8	9	10	11	12	13	comp sys & database
	3	14	15	16	17	18	19	20	comp sys & database
	<i>reading-week</i>	21	22	23	24	25	26	27	
October									
	4	28	29	30	1	2	3	4	comp sys & database
	5	5	6	7	8	9	10	11	comp sys & database
	6	12	13	14	15	16	17	18	comp sys & database
	7	19	20	21	22	23	24	25	comp sys & database
November									
	<i>reading-week</i>	26	27	28	29	30	31	1	
	8	2	3	4	5	6	7	8	comp sys & database
	9	9	10	11	12	13	14	15	comp sys & database
	<i>reading-week</i>	16	17	18	19	20	21	22	
	10	23	24	25	26	27	28	29	comp sys & database
December									
	11	30	1	2	3	4	5	6	comp sys & database
	12	7	8	9	10	11	12	13	comp sys & database
		14	15	16	17	18	19	20	
		21	22	23	24	25	26	27	
January									
		28	29	30	31	1	2	3	

Weekly Webinar Schedule



Expectations

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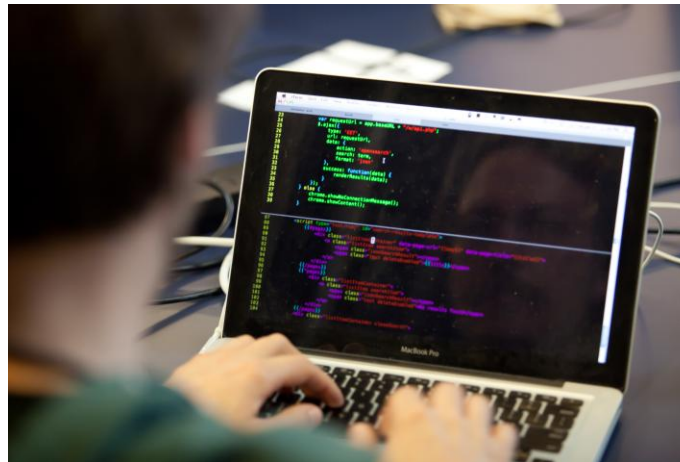
Semester 2 Assessment Schedule

	September			October	November				December				January				
week no.	1	2	3	rd	4	5	6	7	rd	8	9	rd	10	11	12	Rd	
Computer Systems & Networks																	
Assessments				C1					C2								
Database																	
Assessment				D1a					D1b		D2						

TBCA for Caroline's Computer Systems



5% Cisco
NetAcad
chapter exams
(*due end Oct*)



20% Linux Shell scripting

Begins with
5%
Homework



5% InClass Exercises

CA for Caroline's Computer Systems



5% Cisco
NetAcad
chapter exams
(*ongoing*)



Networking Academy

YOU SHOULD HAVE RECEIVED AN EMAIL INVITE FROM ME, SIMILAR TO:



Cisco Networking Academy



Join Linux Essentials at Waterford Institute of Technolo...

Hello

Module 1 - Introduction to Linux

Chapter 1

Module 2 - Operating Systems

Chapter 2

Chapter 02 Exam

Module 3 - Working in Linux

Chapter 3

Chapter 03 Exam

Module 4 - Open Source Software and Licensing

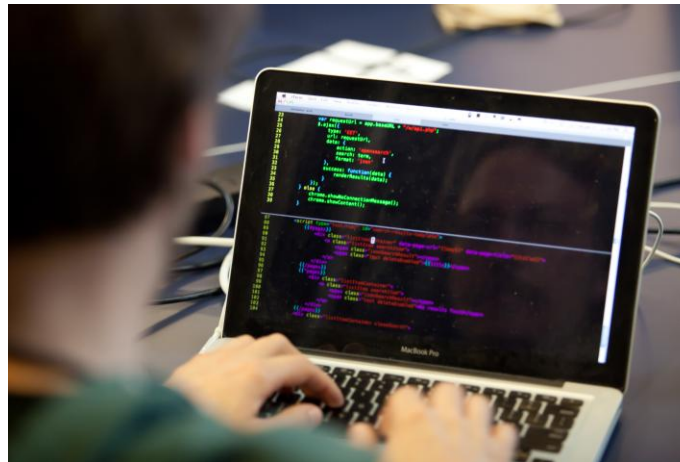
Chapter 4

Chapter 04 Exam

CA for Caroline's Computer Systems



5% Cisco NetAcad
chapter exams (*due
end Oct*)



25% Linux Shell scripting assessment
(Given *end Sept, Due END 2nd Reading
Week*)

Assignments

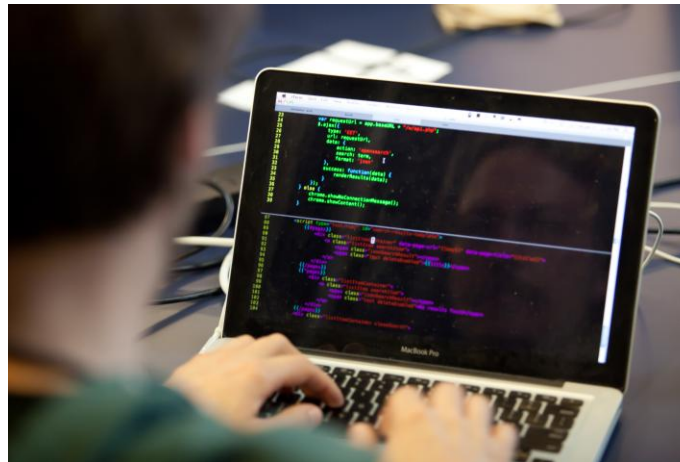
	September			October			November			December				January		
week no.	1	2	3	rd	4	5	6	rd	7	8	9	rd	10	11	12	Rd
Computer Systems & Networks																
Assessments	C1						C2									
Database																
Assessment	D1a						D1b						D2			

2025		S	M	T	W	T	F	S
Semester 2								
September								
1			1	2	3	4	5	6
2	7	8	9	10	11	12	13	
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reading-week	21	22	23	24	25	26	27	
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4	28	29	30	1	2	3	4	
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6	12	13	14	15	16	17	18	
7	19	20	21	22	23	24	25	
November reading-week	26	27	28	29	30	31	1	
8	2	3	4	5	6	7	8	
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reading-week	16	17	18	19	20	21	22	
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11	30	1	2	3	4	5	6	
12	7	8	9	10	11	12	13	
	14	15	16	17	18	19	20	

CA for Caroline's Computer Systems



5% Cisco NetAcad
chapter exams (*due
end Oct*)



25% Linux Shell scripting

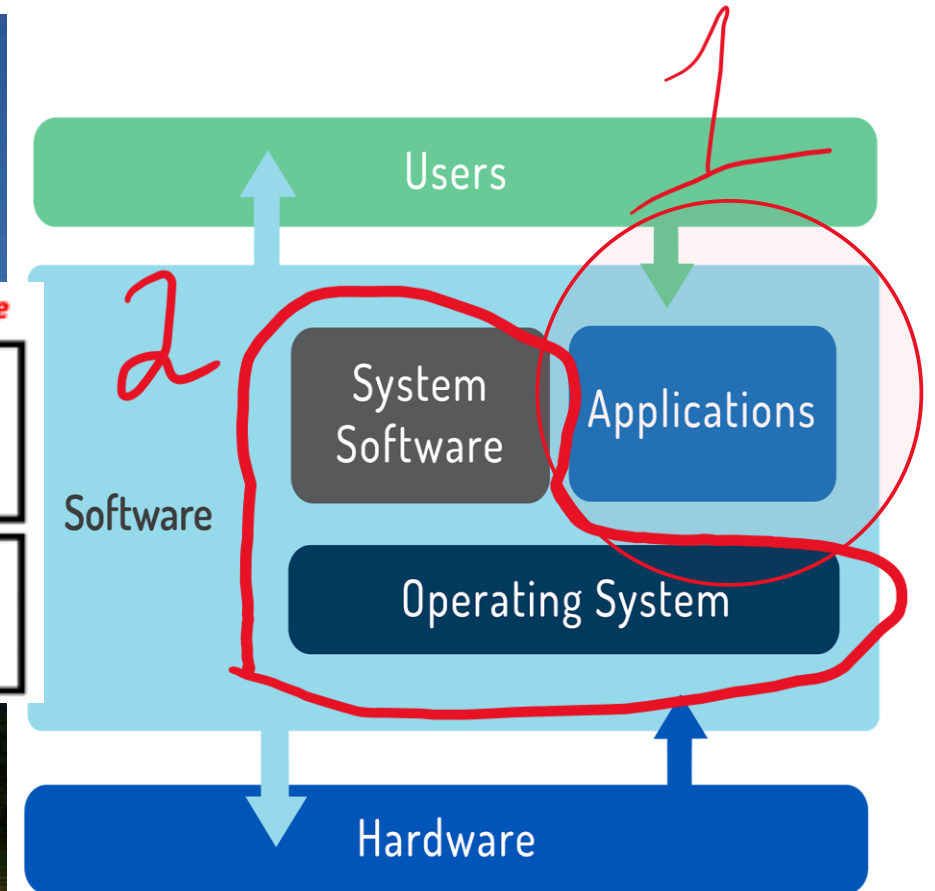
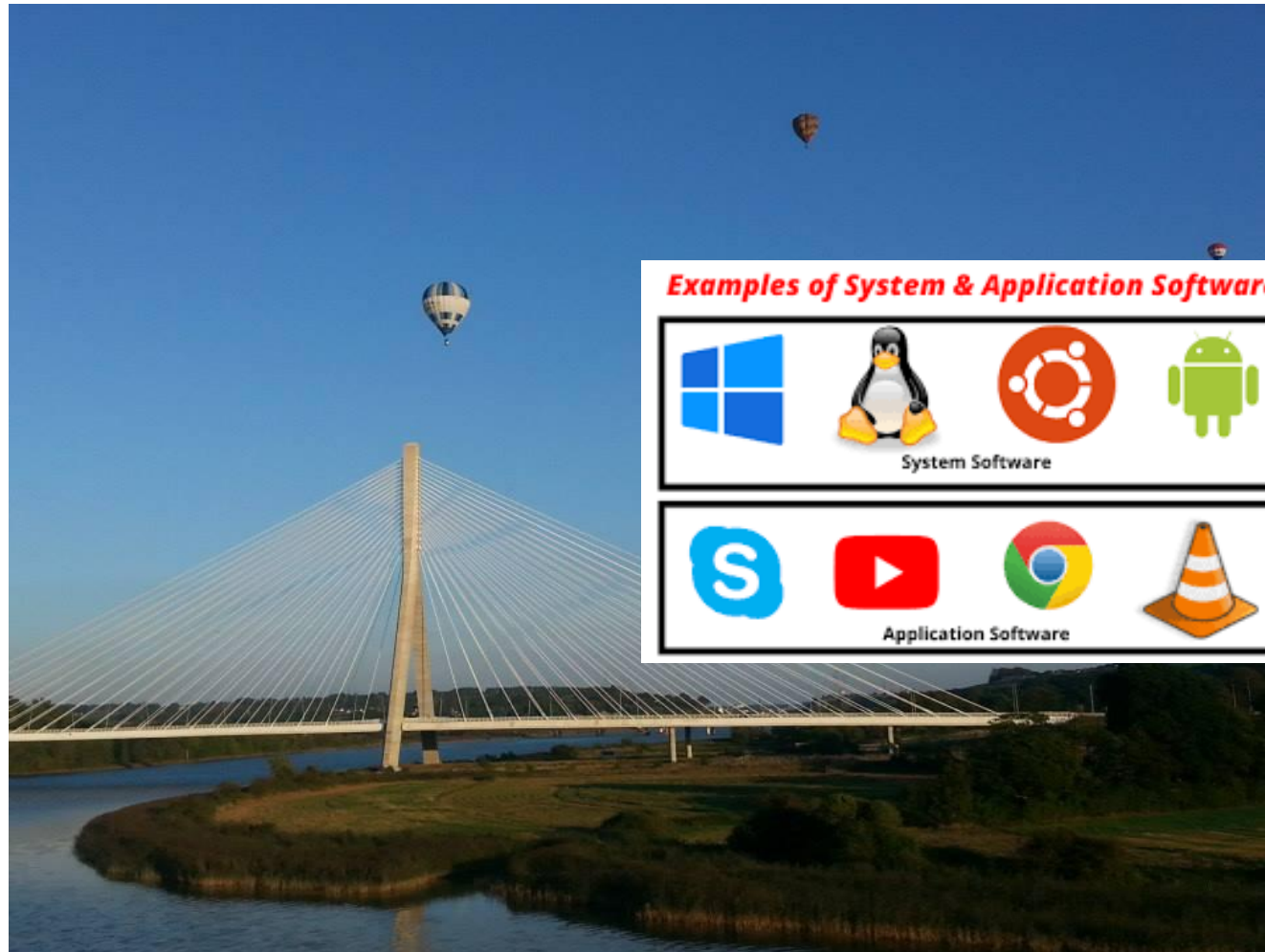


5% Classwork:
completed scripts

Introduction to Operating Systems



Operating System (abbr. OS)



Decision Points when Considering which OS is best

- **Role:** Accessed by 1 user directly (desktop) or many users remotely (server)?
- **Function:** Does it need to run specific software?
- **Life Cycle:** What is the service lifetime?
- **Stability:** Are OS releases *beta* (not tested “in the wild”) or *stable* (tested)?
- **Compatibility:** Is it backwards compatible
- **Cost:** important factor for new systems

Introduction to Linux (as an OS)



Linux is Everywhere!

- Linux is everywhere, Linux jobs are everywhere, there is a shortage of Linux talent, and, Linux skills are in demand in just about every industry and job category.



Default (GPT-3.5)



why is it important to become familiar with shell scripting?



Becoming familiar with shell scripting offers several important benefits, especially for individuals working in IT, software development, data analysis, and system administration. Here are some reasons why it's important to learn shell scripting:

Linux is Open Source

- The development of Linux closely parallels the rise of *open source software*.
- The open source philosophy is that

You have a right to obtain the software **source code** and

You have a right to **modify** it for your own use.

Linux is around a long time

- Linux started in 1991 as a hobby project by Linus Torvalds, a Finnish born computer scientist studying at the University of Helsinki.



Linux is a Kernel

- Linux means the *kernel* of the system, which is the central controller of everything that happens on the computer.
- Linux is a combination of software called **GNU/Linux**, which defines the *operating system*.
 - **GNU** is the *free software* that provides open source equivalents of many common UNIX commands.
 - The **Linux** part of this combination is the *Linux kernel*, which is the core of the operating system.

Linux Embraces the CLI

Two basic types of interfaces available that allow you to interact with the operating system



Linux Distros

A distribution refers to the Linux kernel, tools, and suite of applications that come bundled together.

Linux distros refer to an OS composed of open-source programs designed by numerous developers and programmers.



Lubuntu = Lightweight Ubuntu

- Ubuntu is one of the popular Linux distributions available for desktops and laptops
- A new version of Ubuntu is released every April and October.

Why?

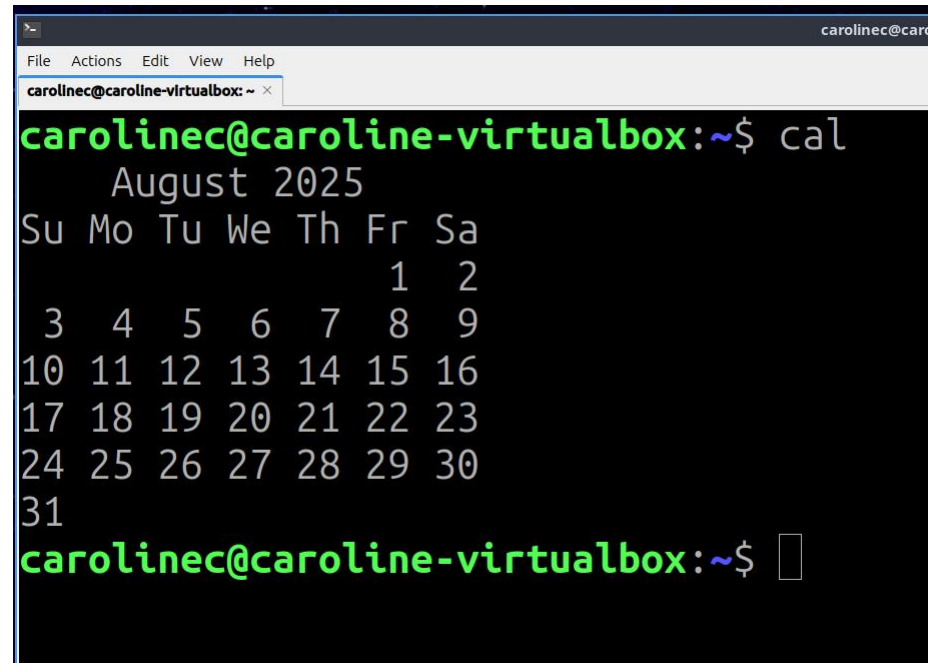
- User-friendly
- There's a thriving Ubuntu community online
- Strong software and hardware support

~~Linux~~ YOU Embrace the CLI

- Enter the Command

\$cal

- to view the built in calendar.



```
carolinec@caroline-virtualbox: ~$ cal
      August 2025
Su Mo Tu We Th Fr Sa
                1  2
 3  4  5  6  7  8  9
10 11 12 13 14 15 16
17 18 19 20 21 22 23
24 25 26 27 28 29 30
31
carolinec@caroline-virtualbox: ~$
```

```
compsys@compsys-virtualbox:~$ cal
Command 'cal' not found, but can be installed with:
sudo apt install ncal
```

\$sudo apt install <<program>>

```
carolinec@caroline-virtualbox:~$ cal Jan 2026
    January 2026
Su Mo Tu We Th Fr Sa
    1  2  3
 4  5  6  7  8  9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30 31

carolinec@caroline-virtualbox:~$
```

```
$sudo apt update
```

Use this command to update the package database any time

~~Linux~~ YOU Embrace the CLI

- Try look up the current weather in Kilkenny using

`$ansiweather -l your_town_name`

```
compsys@compsys-virtualbox:~$ ansiweather -l kilkenny  
Weather in Kilkenny: 18 °C - UVI: 5.32 - Wind: 4.02 m/s WNW - Humidity: 75% - Pressure: 1016 hPa
```

Be familiar with the help **manual** available

What does the `-l` stand for in the command line instruction

`$ansiweather -l kilkenny`



~~Linux~~ YOU Embrace the CLI

- Uninstall ansiweather

\$sudo apt-get remove *program-name*



~~Linux~~ YOU Embrace the CLI

- Try out: `$cowsay Well hello there September!`
- ***Do you need to install it?***
- Once finished, **uninstall**
- Don't like cows?? `$cowsay -l`
`$cowsay -f turkey Almost 100 days to Christmas, gobble!`
- *The `-l` and `-f` are examples of optional arguments that many CLI commands accept*

~~Linux~~ YOU Embrace the CLI

- Try out:

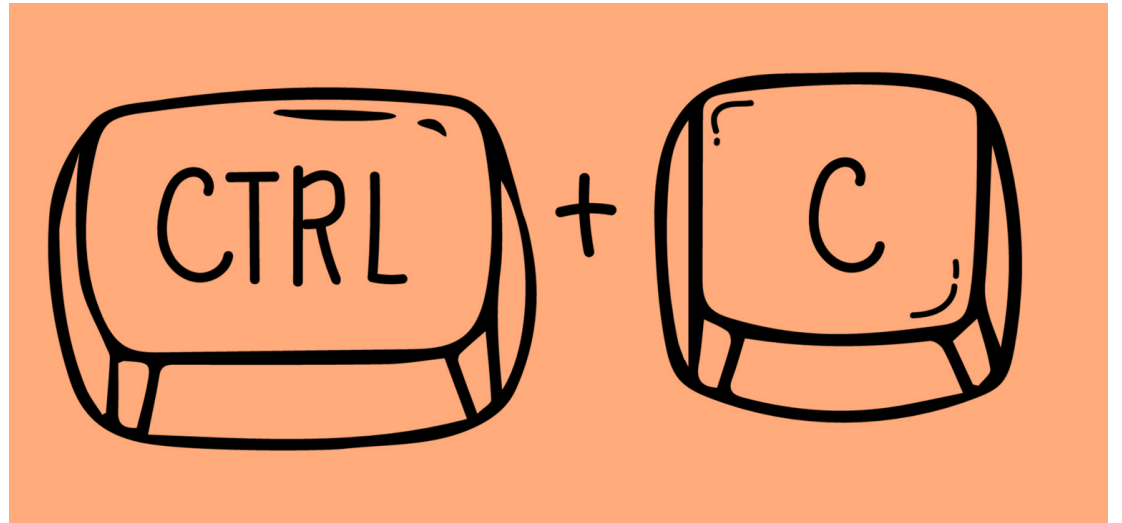
\$cmatrix

Now

\$figlet Lets go

And finally, what happens here
Why?

\$figlet Let's go



Try out this
weeks Lab

01: Computer Systems



NetAcad · Linux as an OS

02: File Permissions



-	rwX	r--	r--
↓	↓	↓	↓
“-” indicates a file “d” indicates directory “l” indicates a link	Read, write, and execute permissions for the owner of the file	Read, write, and execute permissions for members of the group owning the file	Read, write, and execute permissions for other users

Numbering Systems ·
Permissions · Octal

Lab



```
#!/bin/bash
```

Fundamentals · Commands
· Scripts · Permissions